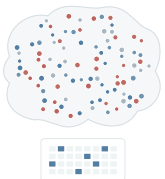
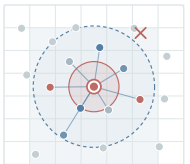
SpatialESS

01



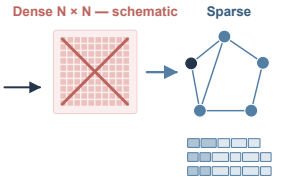
Spatial input

02



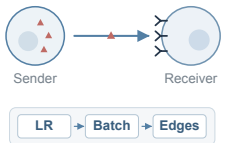
Local neighbors

03



Sparse CSR graph
Spatial graph storage
Dense: O(N²) → CSR: O(N + E)
N: cells; E: stored spatial edges

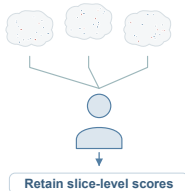
04



Exact LR inference

Reference-compatible LR scoring

05



Patient-level analysis
LMM: condition + (1 | patient)

Original: SpatialCellChat V3

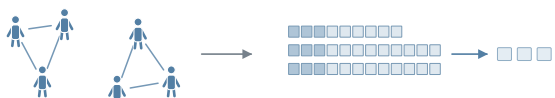


All cell pairs

Materialized cell-pair / LR intermediates

SpatialESS

Icons = cells



Local neighbors

Compact CSR + streaming